

This problem set covers material from Week 5, dates 3/09- 3/11. Unless otherwise noted, all problems are taken from the textbook. Problems can be found at the end of the corresponding subsection.

Instructions: Write or type complete solutions to the following problems and submit answers to the corresponding Canvas assignment. Your solutions should be neatly-written, show all work and computations, include figures or graphs where appropriate, and include some written explanation of your method or process (enough that I can understand your reasoning without having to guess or make assumptions). A general rubric for homework problems appears on the final page of this assignment.

Monday 3/09

1. Prove the following:

- (a) If $T(\mathbf{X}) \equiv T$ is a sufficient statistic for θ and v is a 1:1 function, then $v(T)$ is also sufficient for θ .
 - (b) If $T(\mathbf{X}) \equiv T$ is a sufficient statistic for θ , then the MLE for θ (if it exists) is a function of T .
2. Let $X_1, \dots, X_n | \alpha \stackrel{\text{iid}}{\sim} \text{Gamma}(\alpha, \beta)$ with β known. Demonstrate that the statistic $T = \sum_{i=1}^n \log X_i$ is sufficient for α .

Wednesday 3/11

3. Finish the part of the Rao-Blackwell Theorem we didn't prove in class: $\text{MSE}(\delta^*, \theta) = \text{MSE}(\delta, \theta)$ if only if $\delta^* = \delta$ (where $\delta^* = \mathbb{E}[\delta|T]$ and T is sufficient for θ).

(Note, the “if” direction is trivial, so just need to focus on the “only if” part).

Hopefully you find this next problem interesting! It's broken up into 3 parts.

During World War II, the Allied forces were interested in estimating the total number of tanks produced and owned by the Germans. These tanks were manufactured/labelled with an integer-valued serial number, where each tank was numbered in ascending, consecutive numbers (i.e. the first tank produced was numbered 1, the second tank produced was numbered 2, etc.). Every now and then, after certain battles or by luck, the Allied forces would capture a German tank and thus gain access to the serial number of that tank.

Let N denote the unknown number of tanks produced. At any given time, the Allied forces had n captured tanks in their possession. Let the data $X_1, \dots, X_n \in \{1, 2, \dots, N\}$ be the serial numbers of each captured tank $i = 1, \dots, n$ where i denotes the order of capture. Note that the data are *not* IID Discrete Uniform($\{1, 2, \dots, n\}$)

because once a tank was captured, it was not returned back to the Germans (for obvious reasons). We want to obtain a good estimator of N .

4. **(Part 1 of this problem)** A probably bad estimator would be $\hat{N}_0 = X_1$, the first serial number the Allied forces encountered. Let's try to improve it!
- (a) Find a sufficient statistic $T(\mathbf{X})$ for N . *Remember, the data are NOT IID.*
 - (b) Find the PMF of your sufficient statistic, $P(T = t)$. *Don't forget the support please!*
 - (c) Find the expectation of your sufficient statistic. The following identities might be useful, where x, y are positive integers:
 - $x \binom{x-1}{y-1} = y \binom{x}{y}$
 - $\sum_{k=y}^x \binom{k}{y} = \binom{x+1}{y+1}$
5. **(Part 2 of this problem)** We will continue where we left off in the previous problem.
- (a) Find the conditional PMF of X_1 given your sufficient statistic $T = t$ from the previous problem. *Hint 1: consider the cases when $X_1 = t$ and $X_1 \neq t$. Hint 2: counting/combinatorics...*
 - (b) Use Rao-Blackwell to improve $\hat{N}_0$ to obtain $\hat{N}_{RB}$. Intuitively, does it seem like it really improved $\hat{N}_0$?
 - (c) Demonstrate that your estimator $\hat{N}_{RB}$ is biased for N . Will we tend to over- or under-estimate N ? Then provide an unbiased estimator of N based on your Rao-Blackwellized estimator $\hat{N}_{RB}$. Call it $\hat{N}_{URB}$.
6. **(Part 3 of this problem)** Head over to the provided `.Rmd` template to compare these estimators empirically (i.e. via simulation).

General rubric

Points	Criteria
5	The solution is correct <i>and</i> well-written. The author leaves no doubt as to why the solution is valid.
4.5	The solution is well-written, and is correct except for some minor arithmetic or calculation mistake.
4	The solution is technically correct, but author has omitted some key justification for why the solution is valid. Alternatively, the solution is well-written, but is missing a small, but essential component.
3	The solution is well-written, but either overlooks a significant component of the problem or makes a significant mistake. Alternatively, in a multi-part problem, a majority of the solutions are correct and well-written, but one part is missing or is significantly incorrect.
2	The solution is either correct but not adequately written, or it is adequately written but overlooks a significant component of the problem or makes a significant mistake.
1	The solution is rudimentary, but contains some relevant ideas. Alternatively, the solution briefly indicates the correct answer, but provides no further justification.
0	Either the solution is missing entirely, or the author makes no non-trivial progress toward a solution (i.e. just writes the statement of the problem and/or restates given information).
Notes:	For problems with multiple parts, the score represents a holistic review of the entire problem. Additionally, half-points may be used if the solution falls between two point values above.
Notes:	For problems with code, well-written means only having lines of code that are necessary to solving the problem, as well as presenting the solution for the reader to easily see. It might also be worth adding comments to your code.